

Sarthak Jain

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EDUCATION

University of Pennsylvania

Philadelphia, PA

Master of Science in Computer Science and Data Science, GPA: 4.0/4.0

Expected May 2027

- **Coursework:** Applied Machine Learning, Big Data Analytics, Internet and Web Systems, Operating Systems, Databases
- **Organizations:** Developer and Researcher for AI @ Penn, Claude Builder Club (Placed Top 3 in yearly hackathon)

Rutgers University-New Brunswick

New Brunswick, NJ

Bachelor of Science in Computer Science and Data Science, GPA: 3.91, Major GPA: 3.97

September 2022 - May 2025

- **Coursework:** Data Structures and Algorithms, Discrete Structures, Computer Architecture, Introduction to AI, Deep Learning
- **Organizations:** Backend Developer and Technical Lead at HackRU, ARESTY Research Assistant

WORK EXPERIENCE

Google – Incoming Fall Software Engineer Intern

Sunnyvale, CA

Robinhood – Incoming Summer Software Engineer Intern

Menlo Park, CA

Amazon Web Services (AWS)

New York, NY

Software Development Engineer Intern

May 2025 - August 2025

- Developed an end-to-end system for automated report delivery via console using **Java** for backend, **React + TypeScript** for frontend, and **SQL** for querying, eliminating 50+ manual data requests/month and improving data accessibility for stakeholders
- Built a Redshift ingestion pipeline with real-time S3 triggers, ensuring daily refresh and automated transformation of report data
- Migrated **100K+** daily records from NoSQL **DynamoDB** to **Redshift** using **Lambda**, EventBridge, S3 replication, and SQS
- Created a custom API with presigned URLs to serve 90 days of reports across 10+ report types via a centralized internal console

Machine Learning Lab

New Brunswick, NJ

Undergraduate Researcher

December 2023 – May 2025

- Enhanced LLM evaluation by contributing to the Multitask Language Understanding with Symbol Replacement (MMLU-SR) dataset, showcasing over a 30% drop in model performance, highlighting the symbolic reasoning limitations of different LLMs
- Executed benchmark testing for MMLU-SR on Llama-3-70B and Gemini 1.5 Pro, demonstrating a performance decrease of 25%
- Published evaluation results at peer-reviewed **EMNLP GenBench Workshop** in collaboration with a leading USC research team

Rutgers Rail and Transit Research Lab

New Brunswick, NJ

Undergraduate Researcher

September 2023 – May 2025

- Collaborated on the Using Artificial Intelligence for Next-Generation Intelligent Transportation project that will analyze track safety challenges, path efficiency, and real-time positioning accuracy in 10+ low connection areas of the NJ Transit Rail System
- Applied **OpenCV** and **Python** to generate 3,000+ bounding boxes to validate model accuracy in detecting track safety hazards
- Developed a user-friendly full-stack **Tkinter** interface for crack visualization, video navigation, and defect validation workflows
- Co-authored research paper published in **Smart and Resilient Transportation Journal** on AI applications to enhance rail safety

Gateway (Acquired by Circle)

New York, NY

Software Engineer Intern

June 2024 - August 2024

- Developed protocol layer in a Web3 environment using **Rust**, creating 50+ Zero-Knowledge rollups for anchoring credential data
- Refined **Typescript** SDK to improve developer experience for issuing and consuming verifiable credential data in 7+ languages
- Created 50+ data dashboards to show transaction activity on blockchain protocol layer using Dune Analytics and Flipside Crypto

PROJECTS

Monitord | C++17, CMake, JavaScript, WebSocket, RestAPI

September 2025 – December 2025

- Developed system monitoring server using macOS APIs to collect **10+ metrics** with **1-second precision** and **<5ms overhead**
- Built an asynchronous HTTP/1.1 server with kqueue loop and WebSocket, handling **100+ connections** with **sub-10ms latency**
- Created an interactive web dashboard with Chart.js and WebSocket updates, displaying **8+ metrics** with **60+ data point graphs**
- Designed a time-series storage system with 1-hour rolling window (**3,600+ samples**) and threshold alerts for resource monitoring

Rutgers ML Course Finder | Python, PyTorch, Pinecone, React.js | HackRU 2024 Winner

October 2024

- Implemented Pinecone for optimized storage and **60% faster retrieval** time of vector embeddings, facilitating real-time similarity searches and delivering probabilistic confidence scores for accurate matching of students' interests with course offerings
- Developed website with React.js to search classes using interest-based keyword search and integrated user feedback functionality
- Integrated similarity scoring using **5000+ vector embeddings** created by PyTorch, enabling context-aware retrieval of courses
- Utilized BeautifulSoup4 for web scraping of relevant course information to improve the accuracy and accuracy of search results

TECHNICAL SKILLS

Languages: C++, Java, Python, Go, SQL, R, Rust, Typescript, JavaScript, MATLAB

Technologies: React, AWS, NumPy, Pandas, PyTorch, Pinecone, MySQL, NoSQL, Docker, Git, REST APIs